

Work Experience

Technical Lead: [PETRONAS Digital Sdn Bhd](#), Kuala Lumpur, Malaysia

April 2024 - Current

Project Name: **Datathon**

- Led architecture and execution of **2TB+** RDS-to-Databricks migration, improving analytics speed by **40%** and reducing infra cost by **25%**.
- Directed design of scalable ETL pipelines processing **100K+** daily records with zero data loss and full schema integrity.
- Enabled near real-time analytics by integrating streaming data sources into Databricks, reducing reporting lag from **hours** to under **10 minutes**
- Mentored **4 engineers** and drove cross-team alignment to deliver on-time migration with **100%** SLA compliance.
- Tools & Technologies: Python, PySpark, Delta Lake, AWS, DBC Workflow, Medallion Architecture, Azure DevOps

Project Name: **Heatmap**

- Led a **7-engineer** team to build real-time forecasting for **10+** commodities, enabling insights for **~100** global traders.
- Designed low-latency caching and autoscaling, achieving **P95** latency **<100ms** under peak traffic
- Architected secure CI/CD pipeline using Git, ECR, Lambda, and Secret Manager, ensuring rapid, scalable deployments
- Tools & Technologies: Python, ERC, lambda, CloudWatch, Secret Manager, Docker, SQL, ElastiCache

Software Development Engineer II : [Amazon](#), Vancouver, BC, Canada

June 2022 - August 2023

Project Name: [ElastiCache \(AWS\)](#)

- Worked in the monitoring team to develop and implement automated scaling solutions for nodes, ensuring efficient **scaling up or down** based on demand and prediction
- Designed and completed the Performance Test project for large node (Droplet, **64** vCPUs), sending **100K** concurrent requests by **1000** clients, conducting load testing, monitoring CPUs and memory usage, and implementing automated heat mitigation for optimal performance
- Designed and led integration tests to reach **full coverage**, ensuring seamless functionality and performance benchmark
- Maintaining **Droplet** where cache instances are placed and created dynamic virtual hosts to periodically perform health checks, memory status, I/O, and other related metrics of droplets
- Sketched out and implemented app configuration of the project to modify configurations via commands that retains the project from **redeployment**
- Mentored **two** new joiners to the team, providing guidance, conducting training sessions, offering feedback, and fostering a supportive work environment to ensure seamless integration and professional growth within the organization
- Participate in on-call rotation and diligently followed up on action items for timely resolution
- Tools & Technologies: **Java**, C++, Python, **Spring Boot**, **S3**, **DynamoDB**, **Redis**, **AWS**, **SQL**

Software Engineer: [AnyConnect](#), Dhaka, Bangladesh

January 2020 – May 2022

Project Name: **Smarter AI Camera**

- Worked on the server-side of **microservices** (Report, Repository, Tenant, Geo-location, Auth) with **Java** for providing data during access of camera and dashboard within least possible time
- Introduced **Report Assistant** service to auto-delete outdated data, optimizing database performance and query speed
- Designed, documented and mastered system test for microservices by **python3** language which helped to increase Test coverage from **90%** to **98%**
- Completed system test of ‘*AnyConnect Access Library*’ and ‘*AnyConnect Stream Library(Raspberry PI)*’ with **C++11**
- Tools & Technologies: gRPC, protobuf, Spring Boot, NoSQL(TableStorage), MySQL, ElasticSearch

Software Engineer: [Samsung R&D Institute of Bangladesh](#)

February 2018 – January 2020

Project Name: [VeoSens \(Health Insurance\)](#)

- Introduced mobile site cache which improved **80%** query time for users
- Developed **iBG (iOS BackGround) framework** to continue task when ios app is in background
- Passed Samsung Software Capability(SWC) professional test (top **30%** among **120k** employees all around the globe)
- Mentored **10+** team members in data structures and algorithms, resulting code efficiency across team deliverables

Software Engineer: [Enosis Solutions](#), Dhaka, Bangladesh

August 2017 - February 2018

Project Name: **Bangladesh Thalassemia Foundation Hospital management**

- Implemented online doctor appointment feature on the top of existing system, which helped to gain **20%** more customers than normal and provided on-call support
- Tools & Technologies: ASP.NET Core, SQL Server Reporting Services, Microsoft SQL Server

Teaching Assistant: [Jahangirnagar University](#)

January 2014 – December 2016

- Coached around 100 students for programming competitions and algorithmic problem solving

Certifications

- [AWS Certified](#) Solution [Architect](#) Associate (SAA-C03)

Programming Competitions Experience

- **7th / 100+** position in ACM-ICPC Asia Regional Dhaka 2015 hosted by North South University (Team name: [JU SMS](#))
- Placed in top **3** three times, top **10** seven times in multiple national programming competition events between 2014-17

Education

- M.Sc in Information Technology, Jahangirnagar University, Dhaka, Bangladesh
- B.Sc in Information Technology, Jahangirnagar University, Dhaka, Bangladesh

February 2015 - February 2017
January 2011 - January 2015